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**Azure SQL Database:**

Step-by-step way as a **SQL Server DBA** so you can relate it directly to what you already know from on-prem SQL Server.

### 🌩️ What Is Azure SQL Database?

**Azure SQL Database** is a **Platform as a Service (PaaS)** offering from Microsoft.
It's a **cloud-based version of SQL Server** that removes most of the management overhead (patching, backups, high availability, etc.) while still letting you use **T-SQL** just like in your on-prem SQL Server.

### 🧱 1. Architecture Overview

| Component | Description |
|---|---|
| **Azure SQL Database** | A single database hosted in Azure; similar to a standalone database in SQL Server. |
| **Elastic Pool** | A group of multiple databases that share resources (DTUs/vCores). |
| **Managed Instance** | A PaaS instance that supports **almost full SQL Server feature parity** (Agent, Linked Servers, CLR, etc.). |
| **SQL Server on Azure VM (IaaS)** | Full SQL Server installed on an Azure Virtual Machine; you manage it completely. |

### ✅ Types of Deployments

| Deployment | Description | Best For |
|---|---|---|
| **Single Database** | One isolated database with dedicated resources. | SaaS apps, small/medium workloads. |
| **Elastic Pool** | Multiple DBs sharing compute/resources. | Multi-tenant apps. |
| **Managed Instance** | Closest to full SQL Server instance. | Lift-and-shift migrations, enterprise apps. |
| **SQL Server on VM** | Full control, full responsibility. | Legacy or custom apps needing OS-level access. |

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### ⚙️ 2. Service Tiers & Compute Models

Azure SQL uses **DTU-based** or **vCore-based** purchasing models.

🔷 **DTU Model (Database Transaction Unit)**

- Simplified blend of **CPU + Memory + IOPS**.
- Predefined tiers:
  - Basic
  - Standard (S0–S12)
  - Premium (P1–P15)
- Simpler but less flexible.

🔷 **vCore Model**

- Choose your **CPU, Memory, and Storage** separately.
- Two tiers:
  - **General Purpose**: Best for most workloads. Uses remote storage.
  - **Business Critical**: Local SSDs, best performance and HA.
  - **Hyperscale**: Up to 100 TB, rapid scale-out read replicas.

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| Tier | Storage | Max Size | Use Case |
|---|---|---|---|
| General Purpose | Remote | 4 TB | Most business apps |
| Business Critical | Local SSD | 4 TB | Low-latency OLTP |
| Hyperscale | Remote (multi-tier) | 100 TB | Very large DBs |

### 🔐 3. Security Features

Azure SQL Database offers **enterprise-grade security** out of the box.

| Feature | Description |
|---|---|
| **Firewall Rules** | Control who can connect (server-level & database-level). |
| **Authentication** | SQL Auth, Azure AD, Managed Identity. |
| **Encryption** | TDE (Transparent Data Encryption) enabled by default. |
| **Always Encrypted** | Encrypt sensitive data at column level (client-side). |
| **Auditing & Threat Detection** | Built-in logging & anomaly detection. |
| **Row-Level Security** | Restrict access at row-level. |

### 🔄 4. Backup, Restore, and HA

| Feature | Description |
|---|---|
| **Automated Backups** | Azure automatically takes full, diff, and log backups. |
| **Point-in-Time Restore (PITR)** | Restore to any second within retention (7–35 days default). |
| **Geo-Replication** | Active geo-replica for DR; up to 4 readable secondaries. |
| **Zone Redundant HA** | Built-in high availability across Availability Zones. |
| **Long-term Retention (LTR)** | Store backups up to 10 years in Azure Blob Storage. |

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### 🚀 5. Scaling Options

You can **scale up/down** or **scale out** easily:

| Type | Description |
|---|---|
| **Vertical Scaling** | Increase/decrease compute size (DTUs or vCores). |
| **Horizontal Scaling** | Add replicas for read scale-out (Hyperscale). |
| **Serverless Tier** | Automatically pauses/resumes; pay per second of usage. Ideal for intermittent workloads. |

### 📊 6. Monitoring and Performance

Azure provides several built-in tools:

| Tool | Purpose |
|---|---|
| **Azure Portal Metrics** | View DTU/vCore, CPU, IO utilization, etc. |
| **Query Performance Insight** | Visualize top queries and resource usage. |
| **Automatic Tuning** | Identifies and applies query plan fixes and index recommendations. |
| **Azure Monitor / Log Analytics** | Centralized logging and performance tracking. |
| **Dynamic Management Views (DMVs)** | Same as SQL Server for query troubleshooting. |

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## 🔄 7. Migration Paths (Lift & Shift)

| Source | Tool | Description |
|---|---|---|
| On-Prem SQL Server | **Azure Database Migration Service (DMS)** | Automates schema & data migration. |
| SSMS | Deploy to Azure SQL Wizard | Quick method for small DBs. |
| **BACPAC Import/Export** | For schema + data migration (smaller DBs). | |
| **Managed Instance** | For minimal code change migration. | |

## 💼 8. Common Administration Tasks

| Task | Azure Equivalent |
|---|---|
| Creating DB | CREATE DATABASE or via Portal |
| Restoring DB | Portal → Point-in-Time Restore |
| Jobs | Use **Elastic Jobs** or **Logic Apps** (since SQL Agent isn't in Single DB) |
| Maintenance | Azure handles backups, patching, HA |
| Performance Tuning | Use Query Store, Automatic Tuning, DMVs |

## 💰 9. Cost Optimization Tips

1. Choose **vCore General Purpose + Azure Hybrid Benefit** (reuse your SQL license).
2. Use **Serverless Tier** for low-usage databases.
3. Use **Auto Pause** to save compute cost.
4. Configure **right-size storage** — don't overprovision.
5. Use **Azure Advisor** for performance and cost recommendations.

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## 🧩 10. Key Differences: On-Prem vs Azure SQL Database

| Feature | On-Prem SQL Server | Azure SQL Database |
|---|---|---|
| Infrastructure | You manage | Microsoft manages |
| Backups | Manual | Automatic |
| Patching | Manual | Automatic |
| SQL Agent | Yes | No (use Elastic Jobs) |
| Linked Servers | Yes | Limited / Not supported |
| CLR | Limited | Only in Managed Instance |
| Filegroups | Yes | Not available (abstracted) |

## 🧠 11. Common Real-World DBA Scenarios

### 🧩 Scenario 1: Performance Tuning

Use **Query Store + Performance Insight** to identify top 10 CPU-consuming queries and apply **index tuning** or **plan correction**.

### 🧩 Scenario 2: Database Restore to Specific Time

In Portal → Select DB → **Restore → Point-in-Time Restore** → Select timestamp → New DB created with that state.

### 🧩 Scenario 3: Geo-Replication for DR

Enable **Active Geo-Replication** → Secondary DB in another region → Use for read queries or failover.

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## ⚡ 12. Tools to Manage Azure SQL

- **Azure Portal**
- **SQL Server Management Studio (SSMS)**
- **Azure Data Studio**
- **PowerShell / Azure CLI**
- **T-SQL scripts**

Example:

```
-- Connect to Azure SQL Database
SELECT name, create_date
FROM sys.databases;

-- View current resource usage
SELECT * FROM sys.dm_db_resource_stats;
```

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**Practical Hands-on Steps (Creating DB, Configuring Firewall, Querying, and Scaling)** using Azure Portal + T-SQL examples

**Step-by-step, practically**, just like a live **Azure SQL DBA hands-on session** — from creating your first database in Azure to managing it using SSMS & T-SQL.
Let's begin 🚀

🧭 **Hands-On: Azure SQL Database Practical Setup & Administration**
🧩 **Prerequisites**
✅ Active **Azure Subscription**
✅ **SQL Server Management Studio (SSMS)** or **Azure Data Studio** installed
✅ Basic knowledge of SQL Server

🌐 **Step 1: Create an Azure SQL Database (Single Database)**
**Option A — Using Azure Portal**

1. **Login** to Azure Portal.
2. Search for **"SQL Database"** → Click **+ Create**.
3. Under **Basics**:
   - **Subscription:** Choose your Azure subscription.
   - **Resource Group:** Create or select existing (e.g., RG-SQLDemo).
   - **Database Name:** AdventureWorksAzure
   - **Server:** Create new
     - Server name: sqlserverdemo2025
     - Location: e.g., *East US*
     - Authentication: SQL authentication
     - Login: sqladmin
     - Password: strong password
   - **Compute + Storage:** Choose **General Purpose (vCore)** → 2 vCores, 32 GB storage (for demo).
   - **Backup storage redundancy:** Locally-redundant (default).
4. **Networking Tab:**
   - Connectivity method: **Public Endpoint**
   - Firewall rules: Allow Azure services and your client IP address.
5. **Review + Create** → click **Create**.
   Deployment takes ~2–3 minutes.

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**Option B — Using Azure CLI**
If you prefer CLI:

```
az sql server create \
 --name sqlserverdemo2025 \
 --resource-group RG-SQLDemo \
 --location eastus \
 --admin-user sqladmin \
 --admin-password P@ssword123
```

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```
az sql db create \
  --resource-group RG-SQLDemo \
  --server sqlserverdemo2025 \
  --name AdventureWorksAzure \
  --service-objective GP_Gen5_2
```

🔐 **Step 2: Configure Firewall Rules**

To allow your local SSMS to connect:

1. Go to **SQL Server** → sqlserverdemo2025.
2. Click **Networking** → **Add client IP** → **Save**.
3. This adds your local public IP to the server-level firewall.

🖥️ **Step 3: Connect via SSMS**

1. Open **SQL Server Management Studio**.
2. **Server type:** Database Engine
3. **Server name:** sqlserverdemo2025.database.windows.net
4. **Authentication:** SQL Server Authentication
   - Login: sqladmin
   - Password: (your password)
5. Click **Connect** ✅

🧱 **Step 4: Create Schema & Tables**

Let's create a small sample schema and table for testing.

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```
-- Create a schema
CREATE SCHEMA Sales;

-- Create a table
CREATE TABLE Sales.Orders
(
   OrderID INT IDENTITY(1,1) PRIMARY KEY,
   OrderDate DATETIME DEFAULT GETDATE(),
   CustomerName NVARCHAR(100),
   OrderAmount DECIMAL(10,2)
);

-- Insert sample records
INSERT INTO Sales.Orders (CustomerName, OrderAmount)
VALUES ('John Doe', 250.75),
     ('Mary Smith', 560.50),
     ('Steve Lee', 125.00);

-- Check data
SELECT * FROM Sales.Orders;
```

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### 📊 Step 5: Check Performance and Resource Usage

Azure SQL automatically monitors resource consumption.

**Query real-time performance:**

```
SELECT *
FROM sys.dm_db_resource_stats
ORDER BY end_time DESC;
```

**Check database properties:**

```
SELECT name, create_date, containment_desc, compatibility_level
FROM sys.databases;
```

### 🔄 Step 6: Point-in-Time Restore (Backup & Recovery)

1. Go to your database → **Overview** → **Restore**.
2. Choose **Point-in-Time Restore**.
3. Select the timestamp (within 7–35 days retention).
4. New database will be created (e.g., AdventureWorksAzure-Restore).

No need to manage backup jobs — Azure handles:

- Full backup (every week)
- Differential backup (every 12 hours)
- Transaction log (every 5–10 minutes)

### 🌎 Step 7: Configure Geo-Replication (Disaster Recovery)

1. Go to your database → **Geo-Replication**.
2. Click the map → Choose a secondary region (e.g., *West US*).
3. Click **Add Replica**.
4. A **readable secondary** will be created — you can use it for reporting.

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### 🧠 Step 8: Performance Tuning with Query Store

Azure SQL Database has **Query Store enabled by default**.

```
-- View top CPU-consuming queries
SELECT TOP 10
   q.query_id,
   q.query_text_id,
   t.query_sql_text,
   rs.avg_duration,
   rs.avg_cpu_time
FROM sys.query_store_query q
JOIN sys.query_store_query_text t ON q.query_text_id = t.query_text_id
JOIN sys.query_store_runtime_stats rs ON q.query_id = rs.query_id
ORDER BY rs.avg_cpu_time DESC;
```

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### ⚙️ Step 9: Scale Up / Down (Compute Tier)

**Portal:**

→ Go to Database → Compute + Storage → Configure → Change vCores or Tier → Apply

**T-SQL (via PowerShell/CLI):**

```
az sql db update \
 --resource-group RG-SQLDemo \
 --server sqlserverdemo2025 \
 --name AdventureWorksAzure \
 --service-objective GP_Gen5_4
```

🕐 Scaling takes a few minutes but **no data loss**.

### 🧩 Step 10: Automatic Tuning

Automatic Tuning helps optimize performance automatically.

- Go to DB → **Automatic tuning**
  - Enable **Create Indexes**
  - Enable **Drop Indexes**
  - Enable **Force Last Good Plan**

This allows Azure to detect and fix slow query plans.

### 📜 Step 11: Cost Optimization Settings

- Configure **Auto Pause (Serverless Tier)** for idle databases:
- az sql db update \
- --resource-group RG-SQLDemo \
- --server sqlserverdemo2025 \
- --name AdventureWorksAzure \
- --compute-model Serverless \
- --auto-pause-delay 60

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👉 Database pauses after 60 min inactivity and resumes automatically.

- Enable **Azure Hybrid Benefit** to use your existing SQL licenses.

### 🔒 Step 12: Security Best Practices

| Security Measure | Description |
|---|---|
| **TDE (Transparent Data Encryption)** | Enabled by default. |
| **Always Encrypted** | Protects sensitive columns (e.g., SSN). |
| **Auditing** | Store logs in Azure Storage. |
| **Vulnerability Assessment** | Built-in scan for security issues. |
| **Private Endpoint** | Access DB securely within VNet (no public IP). |

Example:

```
SELECT * FROM sys.dm_database_encryption_keys;
```

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## 📘 Summary — What You've Learned

| Topic | Key Takeaway |
|---|---|
| Create DB | Azure Portal / CLI |
| Firewall | Allow local client IP |
| Querying | SSMS same as SQL Server |
| Backup & Restore | Automatic & PITR |
| Geo-Replication | DR in another region |
| Performance | Query Store + Automatic Tuning |
| Scaling | On-demand vertical/horizontal scaling |
| Security | Encryption, Auditing, and Firewall Rules |

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